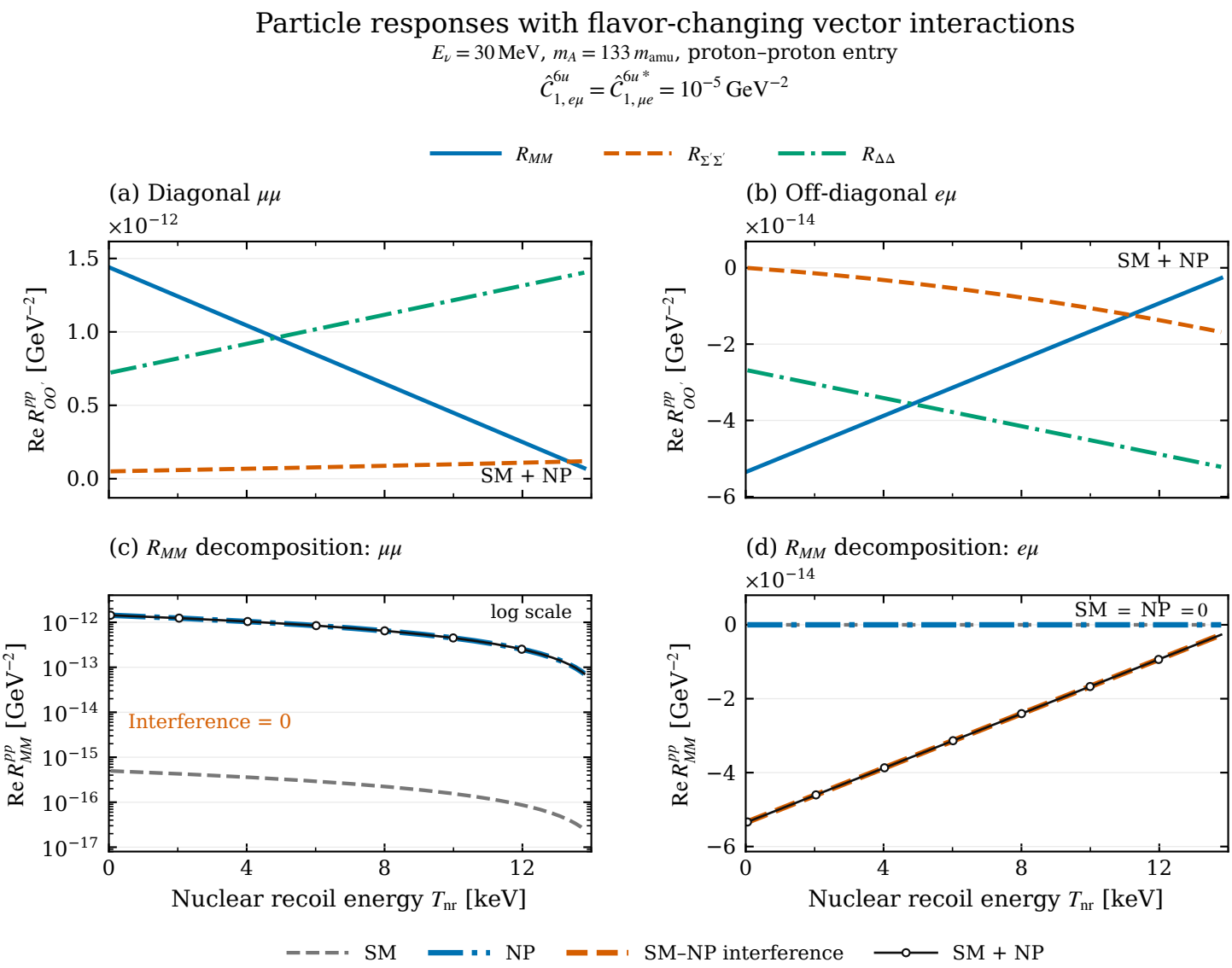

Particle responses with flavor-changing vector interactions
$E_\nu = 30\,\mathrm{MeV}$, $m_A = 133\,m_{\mathrm{amu}}$, proton–proton entry
$\hat{C}^{6u}_{1,e\mu} = \hat{C}^{6u\,*}_{1,\mu e} = 10^{-5}\,\mathrm{GeV}^{-2}$
R_{MM}
$R_{\Sigma'\Sigma'}$
$R_{\Delta\Delta}$
(a) Diagonal $\mu\mu$
$\times 10^{-12}$
$\mathrm{Re}\,R^{pp}_{OO'}$ [GeV^{-2}]
SM + NP
(b) Off-diagonal $e\mu$
$\times 10^{-14}$
$\mathrm{Re}\,R^{pp}_{OO'}$ [GeV^{-2}]
SM + NP
(c) R_{MM} decomposition: $\mu\mu$
log scale
Interference = 0
$\mathrm{Re}\,R^{pp}_{MM}$ [GeV^{-2}]
Nuclear recoil energy T_{nr} [keV]
(d) R_{MM} decomposition: $e\mu$
$\times 10^{-14}$
SM = NP = 0
$\mathrm{Re}\,R^{pp}_{MM}$ [GeV^{-2}]
Nuclear recoil energy T_{nr} [keV]
SM
NP
SM–NP interference
SM + NP